Jacobians and Abel–Jacobi Maps

Let X be a compact connected Riemann surface of genus $g \geq 0$, with base point x_0 and its complex orientation. Homology is singular homology, $U^\vee = \mathrm{Hom}_{\mathbf{C}}(U, \mathbf{C})$, and homomorphisms of complex tori are holomorphic group homomorphisms. Put

$$W = H^0(X, \Omega^1_X), \qquad V = W^\vee.$$

1 Homology

Proposition 1.1. *There is a presentation*

$$\pi_1(X, x_0) \cong \left\langle a_1, b_1, \ldots, a_g, b_g \;\middle|\; \prod_{j=1}^{g} [a_j, b_j] = 1 \right\rangle, \qquad [a, b] = aba^{-1}b^{-1}. \tag{1.1}$$

The Hurewicz map h induces a canonical isomorphism

$$\pi_1(X, x_0)^{\mathrm{ab}} \xrightarrow{\sim} H_1(X, \mathbf{Z}), \qquad H_1(X, \mathbf{Z}) \cong \mathbf{Z}^{2g}.$$

The homology classes of the generators form a symplectic basis:

$$a_i \cdot a_j = b_i \cdot b_j = 0, \qquad a_i \cdot b_j = \delta_{ij}. \tag{1.2}$$

Moreover, $\pi_1(X, x_0)$ is trivial for $g = 0$, is $\mathbf{Z}^2$ for $g = 1$, and is nonabelian for $g \geq 2$.

Proof. The polygon decomposition has one vertex, $2g$ edges, and one face attached by $\prod_j [a_j, b_j]$; van Kampen gives (1.1). For $g = 0$ use the sphere with one vertex and one face. If F is the free group on the edges and N is the normal closure of the relator, then $N \subseteq [F, F]$, so

$$(F/N)^{\mathrm{ab}} = F/(N[F, F]) = F/[F, F] \cong \mathbf{Z}^{2g}.$$

Degree-one Hurewicz identifies this quotient with $H_1(X, \mathbf{Z})$. The standard oriented edge cycles have intersections (1.2). For $g = 1$ the relation is commutativity; for $g \geq 2$ the quotient obtained by setting every $b_j = 1$ is the nonabelian free group on $a_1, \ldots, a_g$. □

2 Periods

Choose a compatible Riemannian metric and use $*dx = dy$, $*dy = -dx$ in a holomorphic coordinate $z = x + iy$. Set

$$\mathcal{H}^1(X, \mathbf{R}) = \{\alpha \in \Omega^1(X, \mathbf{R}) : d\alpha = d * \alpha = 0\}.$$

Theorem 2.1 (Hodge)**.** *Every real de Rham class has a unique harmonic representative; thus $\mathcal{H}^1(X, \mathbf{R}) \xrightarrow{\sim} H^1_{\mathrm{dR}}(X, \mathbf{R})$.*

Lemma 2.2. *The real-linear map*

$$W_{\mathbf{R}} \xrightarrow{\sim} \mathcal{H}^1(X, \mathbf{R}), \qquad \omega \longmapsto \operatorname{Re}\omega$$

has inverse $\alpha \mapsto \alpha + i * \alpha$. *Consequently,*

$$\dim_{\mathbf{C}} W = g, \qquad H^1_{\mathrm{dR}}(X, \mathbf{C}) \cong W \oplus \overline{W}. \tag{2.1}$$

Proof. For $\alpha = p\,dx + q\,dy$,

$$\alpha + i * \alpha = (p - iq)\,dz, \qquad d\alpha = d * \alpha = 0 \iff q_x - p_y = p_x + q_y = 0.$$

These are the Cauchy–Riemann equations for $p - iq$. Conversely, the real part of a holomorphic one-form satisfies these equations and recovers the form by $\alpha + i * \alpha$. Hodge theory and de Rham's theorem give $2\dim_{\mathbf{C}} W = \dim_{\mathbf{R}} H^1(X, \mathbf{R}) = 2g$; complexifying the harmonic representatives gives the decomposition in (2.1). □

Holomorphic one-forms are closed. Integration therefore defines

$$P : H_1(X, \mathbf{Z}) \longrightarrow V, \qquad P(c)(\omega) = \int_c \omega, \tag{2.2}$$

and its real-linear extension $P_{\mathbf{R}} : H_1(X, \mathbf{R}) \to V_{\mathbf{R}}$.

Definition 2.3. A *full lattice* in a real vector space is the integral span of a real basis. A *complex torus* is a quotient U/Γ, where U is a finite-dimensional complex vector space and Γ is a full lattice in $U_{\mathbf{R}}$.

Theorem 2.4. *The map* $P_{\mathbf{R}}$ *is an isomorphism. Hence* P *is injective and*

$$\Lambda_X = P(H_1(X, \mathbf{Z}))$$

is a full lattice of rank $2g$ *in* V.

Proof. If $P_{\mathbf{R}}(c) = 0$, then

$$\int_c \operatorname{Re}\omega = \operatorname{Re}\int_c \omega = 0 \qquad (\omega \in W).$$

By Lemma 2.2 and Hodge theory, these real parts represent all of $H^1_{\mathrm{dR}}(X, \mathbf{R})$. Nondegeneracy of the de Rham pairing gives $c = 0$. Both spaces have real dimension $2g$, so $P_{\mathbf{R}}$ is an isomorphism. If $c_1, \ldots, c_{2g}$ is an integral homology basis, then $P(c_1), \ldots, P(c_{2g})$ is a real basis of $V_{\mathbf{R}}$ and

$$\Lambda_X = \bigoplus_{j=1}^{2g} \mathbf{Z} P(c_j).$$

Thus Λ_X is discrete, and the compact parallelepiped $\{\sum_j t_j P(c_j) : 0 \le t_j \le 1\}$ surjects onto V/Λ_X. □

Definition 2.5. The *Jacobian* of X is

$$J(X) = V/\Lambda_X = H^0(X, \Omega^1_X)^\vee / P(H_1(X, \mathbf{Z})). \tag{2.3}$$

Lattice translations act freely and properly discontinuously by biholomorphisms. Hence $J(X)$ is a compact complex Lie group of complex dimension g, and $J(X) \cong \mathbf{R}^{2g}/\mathbf{Z}^{2g}$ as real manifolds after choosing an integral homology basis.

3 Abel–Jacobi Maps

Definition 3.1. For $x \in X$, choose a piecewise smooth path c from x_0 to x and set

$$\mathrm{AJ}_{x_0}(x) = \left[\omega \longmapsto \int_c \omega\right] \in J(X). \tag{3.1}$$

Proposition 3.2. *The map AJ_{x_0} is well defined and holomorphic, with $\mathrm{AJ}_{x_0}(x_0) = 0$. Under $T_{\mathrm{AJ}_{x_0}(x)}J(X) = V$,*

$$d(\mathrm{AJ}_{x_0})_x(v)(\omega) = \omega_x(v). \tag{3.2}$$

For another base point y_0,

$$\mathrm{AJ}_{y_0}(x) = \mathrm{AJ}_{x_0}(x) - \mathrm{AJ}_{x_0}(y_0).$$

Proof. Two choices of c differ by an integral cycle, so their integral functionals differ by an element of Λ_X. On a coordinate disk, holomorphic primitives give a holomorphic local lift to V and imply (3.2). Concatenation of paths gives the base-point formula. □

Lemma 3.3. *For a complex torus $T = U/\Gamma$, the quotient $q : U \to T$ is a universal cover. Lifting based loops from 0 defines an isomorphism*

$$\delta_T : \pi_1(T, 0) \xrightarrow{\sim} \Gamma, \qquad [\gamma] \longmapsto \widetilde{\gamma}(1).$$

Proof. The lattice action gives a covering and U is contractible. Endpoints of lifted concatenations add. Each $\lambda \in \Gamma$ is realized by $t \mapsto q(t\lambda)$, and a lift with endpoint 0 is a loop in the simply connected space U. □

Theorem 3.4. *Writing $\delta = \delta_{J(X)}$, one has*

$$\delta \circ (\mathrm{AJ}_{x_0})_* = P \circ h. \tag{3.3}$$

Thus $(\mathrm{AJ}_{x_0})_*$ is surjective and

$$\ker((\mathrm{AJ}_{x_0})_*) = [\pi_1(X, x_0), \pi_1(X, x_0)], \qquad \pi_1(J(X), 0) \cong \pi_1(X, x_0)^{\mathrm{ab}}.$$

Moreover, AJ_{x_0} induces an isomorphism $H_1(X, \mathbf{Z}) \xrightarrow{\sim} H_1(J(X), \mathbf{Z})$.

Proof. For a based loop $\gamma : [0, 1] \to X$, the lift of $\mathrm{AJ}_{x_0} \circ \gamma$ starting at zero is

$$\widetilde{\gamma}(t)(\omega) = \int_{\gamma|_{[0,t]}} \omega.$$

Its endpoint is $P(h([\gamma]))$, proving (3.3). The assertions about fundamental groups follow from injectivity of P and degree-one Hurewicz. Abelianizing (3.3) and using naturality of Hurewicz proves the homology assertion. □

4 Universal Property

Lemma 4.1. *For complex tori $T_i = U_i/\Gamma_i$,*

$$\mathrm{Hom}(T_1, T_2) \cong \{A \in \mathrm{Hom}_{\mathbf{C}}(U_1, U_2) : A(\Gamma_1) \subseteq \Gamma_2\}.$$

Also, $H^0(T_i, \Omega^1_{T_i}) = U_i^\vee$ via invariant one-forms.

Proof. A homomorphism lifts uniquely to a holomorphic $A : U_1 \to U_2$ with $A(0) = 0$. The continuous map $(u, v) \mapsto A(u + v) - A(u) - A(v)$ takes values in Γ_2 and vanishes at $(0, 0)$, so A is additive. Hence $dA_u = dA_0$ and $A(u) = dA_0(u)$ is complex-linear. It maps Γ_1 into Γ_2. Conversely, such a linear map descends to a homomorphism. The coefficients of the lift of a holomorphic one-form to U_i are entire and Γ_i-periodic; they are bounded and thus constant by Liouville's theorem. □

Theorem 4.2. *Under $H^0(J(X), \Omega^1_{J(X)}) = V^\vee = W$, the map*

$$\mathrm{AJ}^*_{x_0} : H^0(J(X), \Omega^1_{J(X)}) \xrightarrow{\sim} W$$

is the identity. For every complex torus T and holomorphic map $f : X \to T$, there is a unique homomorphism $F : J(X) \to T$ such that

$$f(x) = f(x_0) + F(\mathrm{AJ}_{x_0}(x)). \tag{4.1}$$

Proof. The assertion on one-forms follows from (3.2). After translating f, assume $f(x_0) = 0$ and write $T = U/\Gamma$. Dualizing pullback of invariant forms gives

$$A = (f^*)^\vee : V \longrightarrow U, \qquad \ell(A(v)) = v(f^*\ell) \quad (\ell \in U^\vee).$$

For a path c from x_0, let u_c be the endpoint of the lift of $f \circ c$ to U starting at zero. Then

$$\ell\left(A\left(\omega \longmapsto \int_c \omega\right)\right) = \int_c f^*\ell = \ell(u_c).$$

For a loop, $u_c \in \Gamma$. Since h is surjective, this implies $A(\Lambda_X) \subseteq \Gamma$. Lemma 4.1 gives a homomorphism $F : J(X) \to T$, and the same identity for an arbitrary path proves (4.1). If F_1, F_2 give the factorization, their induced maps on fundamental groups agree after composition with $(\mathrm{AJ}_{x_0})_*$. This map is surjective by Theorem 3.4, so the linear lifts of F_1, F_2 agree on Λ_X. The lattice spans $V_{\mathbf{R}}$, hence the lifts and the homomorphisms agree. □

5 Polarization

Assume $g \geq 1$ and choose the symplectic basis (1.2).

Lemma 5.1. *For closed complex one-forms α, β,*

$$\int_X \alpha \wedge \beta = \sum_{j=1}^{g} \left(\int_{a_j} \alpha \int_{b_j} \beta - \int_{b_j} \alpha \int_{a_j} \beta \right). \tag{5.1}$$

For $0 \neq \omega \in W$,

$$\frac{i}{2} \int_X \omega \wedge \overline{\omega} > 0. \tag{5.2}$$

Proof. If u_j, v_j are the cohomology classes dual to a_j, b_j, Poincaré duality gives $\int_X u_i \smile u_j = \int_X v_i \smile v_j = 0$ and $\int_X u_i \smile v_j = \delta_{ij}$. Expanding $[\alpha], [\beta]$ gives (5.1). Locally, $\frac{i}{2}\omega \wedge \overline{\omega} = |f(z)|^2 dx \wedge dy$ for $\omega = f(z)\, dz$, proving (5.2). □

Proposition 5.2. *There is a unique basis $\omega_1, \ldots, \omega_g$ of W satisfying $\int_{a_j} \omega_i = \delta_{ij}$. For*

$$\tau_{ij} = \int_{b_j} \omega_i, \qquad Y = \operatorname{Im} \tau,$$

one has $\tau^{\mathsf{T}} = \tau$ and $Y > 0$. In the coordinates $v \mapsto (v(\omega_1), \ldots, v(\omega_g))^{\mathsf{T}}$,

$$\Lambda_X = \mathbf{Z}^g + \tau \mathbf{Z}^g, \qquad J(X) \cong \mathbf{C}^g / (\mathbf{Z}^g + \tau \mathbf{Z}^g). \tag{5.3}$$

Proof. If all a-periods of ω vanish, (5.1) gives $\int_X \omega \wedge \overline{\omega} = 0$; thus $\omega = 0$ by (5.2). Since $\dim_{\mathbf{C}} W = g$, the a-period map $W \to \mathbf{C}^g$ is an isomorphism. This gives the normalized basis and (5.3). Since $\omega_i \wedge \omega_j = 0$, (5.1) gives $0 = \tau_{ji} - \tau_{ij}$. For $\omega = \sum_i c_i \omega_i$ it gives

$$\frac{i}{2} \int_X \omega \wedge \overline{\omega} = \overline{c}^{\mathsf{T}} Y c > 0 \qquad (c \neq 0),$$

so Y is positive definite. □

Definition 5.3. A *Riemann form* on a complex torus U/Γ is an integral alternating form E on Γ whose real-bilinear extension satisfies

$$E(iu, iv) = E(u, v), \qquad E(u, iu) > 0 \quad (u \neq 0).$$

It defines a polarization, which is *principal* if $\Gamma \to \mathrm{Hom}_{\mathbf{Z}}(\Gamma, \mathbf{Z})$, $\lambda \mapsto E(\lambda, -)$, is an isomorphism.

Theorem 5.4 (Riemann's criterion)**.** *A complex torus is projective if and only if it admits a Riemann form. A projective complex torus is an abelian variety over* $\mathbf{C}$.

Theorem 5.5. *The form*

$$E(z, w) = \mathrm{Im}(\overline{z}^{\mathsf{T}} Y^{-1} w)$$

defines a canonical principal polarization on $J(X)$, characterized by

$$E(P(c), P(d)) = c \cdot d \qquad (c, d \in H_1(X, \mathbf{Z})). \tag{5.4}$$

Thus $J(X)$ is a principally polarized abelian variety.

Proof. The positive definite Hermitian form $H(z, w) = \overline{z}^{\mathsf{T}} Y^{-1} w$ is conjugate-linear in its first variable. Hence E is alternating, $E(iz, iw) = E(z, w)$, and $E(z, iz) = H(z, z) > 0$ for $z \neq 0$. Writing $\tau = B + iY$, with $B^{\mathsf{T}} = B$, gives

$$\begin{aligned} E(m + \tau n, m' + \tau n') &= (m + Bn)^{\mathsf{T}} n' - n^{\mathsf{T}}(m' + Bn') \\ &= m^{\mathsf{T}} n' - n^{\mathsf{T}} m' \qquad (m, n, m', n' \in \mathbf{Z}^g). \end{aligned}$$

Thus E is integral and has matrix $\mathsf{S} = \left(\begin{smallmatrix} 0 & I_g \\ -I_g & 0 \end{smallmatrix}\right)$ in the period basis. This matrix is unimodular, so the polarization is principal. The same computation gives (5.4), which also proves independence of the symplectic basis. Riemann's criterion gives projectivity. □

6 Endomorphisms

Let $T = U/\Gamma$ have complex dimension g, and let $\mathcal{I}$ denote multiplication by i on $U_{\mathbf{R}} = \Gamma \otimes_{\mathbf{Z}} \mathbf{R}$.

Proposition 6.1. *Restriction to Γ identifies the endomorphism ring with*

$$\begin{aligned} \mathrm{End}(T) &\cong \{A \in \mathrm{End}_{\mathbf{C}}(U) : A(\Gamma) \subseteq \Gamma\} \\ &\cong \{u \in \mathrm{End}_{\mathbf{Z}}(\Gamma) : u_{\mathbf{R}} \mathcal{I} = \mathcal{I} u_{\mathbf{R}}\} \ \subseteq \ \mathrm{End}_{\mathbf{Z}}(\Gamma). \end{aligned}$$

After choosing a lattice basis, the last ring is $M_{2g}(\mathbf{Z})$. In particular, $\mathrm{End}(T)$ *is a free abelian group of finite rank at most $4g^2$.*

Proof. The first description is Lemma 4.1. A real-linear map is complex-linear exactly when it commutes with $\mathcal{I}$. Since Γ spans $U_{\mathbf{R}}$, restriction is injective. A subgroup of the free abelian group $M_{2g}(\mathbf{Z})$ is free of rank at most $4g^2$. □

Proposition 6.2. *For a principally polarized torus (T, E), its polarization-preserving automorphisms correspond, in a symplectic lattice basis, to*

$$\mathrm{Aut}(T, E) \cong \{M \in M_{2g}(\mathbf{Z}) : M_{\mathbf{R}}\mathcal{I} = \mathcal{I}M_{\mathbf{R}},\ M^{\mathsf{T}}\mathsf{S}M = \mathsf{S}\}.$$

For each $n \in \mathbf{Z}$, multiplication by n belongs to $\mathrm{End}(T)$ *and satisfies $E(n\lambda, n\mu) = n^2 E(\lambda, \mu)$.*

Proof. The first equation is the complex-linearity condition in Proposition 6.1; the second is preservation of E. The latter gives an integral inverse $M^{-1} = \mathsf{S}^{-1}M^{\mathsf{T}}\mathsf{S}$, so the induced map is an automorphism. The assertion about multiplication follows from bilinearity. □

Example 6.3. For $\operatorname{Im}\tau > 0$, set $E_\tau = \mathbf{C}/(\mathbf{Z} + \tau\mathbf{Z})$. Then

$$\mathrm{End}(E_\tau) = \{\alpha \in \mathbf{C} : \alpha(\mathbf{Z} + \tau\mathbf{Z}) \subseteq \mathbf{Z} + \tau\mathbf{Z}\},$$

and

$$\mathrm{End}(E_\tau) \neq \mathbf{Z} \iff \tau \text{ is imaginary quadratic over } \mathbf{Q}.$$

In particular,

$$\mathrm{End}(E_i) = \mathbf{Z}[i], \qquad \mathrm{End}(E_{i\sqrt[4]{2}}) = \mathbf{Z}, \qquad \pi_1(E_i) \cong \pi_1(E_{i\sqrt[4]{2}}) \cong \mathbf{Z}^2.$$

Proof. A complex-linear endomorphism of $\mathbf{C}$ is multiplication by a scalar α. The lattice condition is equivalent to

$$\alpha = a + b\tau, \qquad \alpha\tau = c + d\tau, \qquad a, b, c, d \in \mathbf{Z},$$

and hence to

$$b\tau^2 + (a - d)\tau - c = 0. \tag{6.1}$$

If τ is not imaginary quadratic, the polynomial must vanish identically: $b = c = 0$, $a = d$, so $\alpha \in \mathbf{Z}$. Conversely, if $A\tau^2 + B\tau + C = 0$ with integers $A \neq 0, B, C$, then multiplication by $A\tau$ preserves the lattice and is not integer multiplication. Thus the exceptional parameters form a countable set. For $\tau = i$, the lattice is $\mathbf{Z}[i]$ and its scalar multiplier ring is $\mathbf{Z}[i]$. For $\tau = i\sqrt[4]{2}$, (6.1) gives $a = d$ and $b\sqrt{2} + c = 0$, so $b = c = 0$. □

7 Low Genus

Proposition 7.1. *If $g = 0$, then $J(X)$ is a point. If $g = 1$, then $\mathrm{AJ}_{x_0} : X \to J(X)$ is a biholomorphism. If $g \geq 2$, then AJ_{x_0} is not a homotopy equivalence.*

Proof. For $g = 0$, $W = 0$. For $g = 1$, Theorem 4.2 implies that AJ_{x_0} is nonconstant, so it is a surjective holomorphic map of positive degree. By Theorem 3.4, its pullback on integral H^1 is an isomorphism. Choose $\alpha, \beta \in H^1(J(X), \mathbf{Z})$ with $\langle \alpha \smile \beta, [J(X)] \rangle = 1$. Their pullbacks form an integral basis of $H^1(X, \mathbf{Z})$, so

$$\deg(\mathrm{AJ}_{x_0}) = \langle \mathrm{AJ}_{x_0}^* \alpha \smile \mathrm{AJ}_{x_0}^* \beta, [X] \rangle = \pm 1.$$

Positivity gives degree one. Positive local multiplicities then imply that the map is bijective and unramified, hence biholomorphic. For $g \geq 2$, $\pi_1(X, x_0)$ is nonabelian whereas $\pi_1(J(X), 0) \cong \mathbf{Z}^{2g}$. □

For $X = E_\tau$ and $x_0 = [0]$, the form dz identifies V with $\mathbf{C}$; its period lattice is $\mathbf{Z} + \tau\mathbf{Z}$ and $\mathrm{AJ}_0([z]) = [z]$.

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